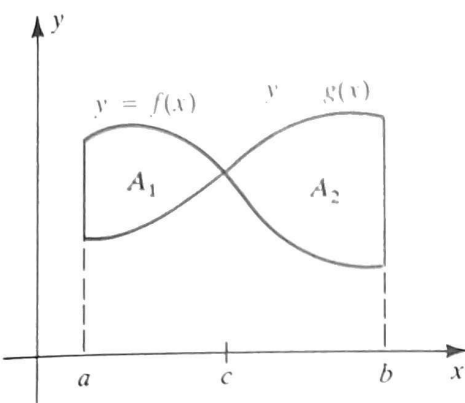


FIGURE 6.16



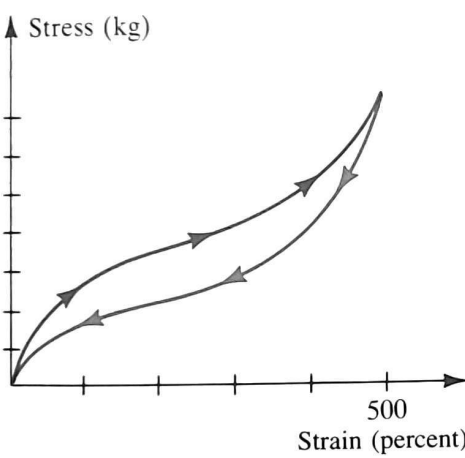
Throughout this section we have assumed that the graphs of the functions (or equations) do not cross one another in the interval under discussion. If the graphs of f and g cross at one point $P(c, d)$, with $a < c < b$, and we wish to find the area bounded by the graphs from $x = a$ to $x = b$, then the methods developed in this section may still be used; however, two integrations are required, one corresponding to the interval $[a, c]$ and the other to $[c, b]$. This is illustrated in Figure 6.16, with $f(x) \geq g(x)$ on $[a, c]$ and $g(x) \geq f(x)$ on $[c, b]$. The area A is given by

$$A = A_1 + A_2 = \int_a^c [f(x) - g(x)] dx + \int_c^b [g(x) - f(x)] dx.$$

If the graphs cross several times, then several integrals may be necessary. Problems in which graphs cross one or more times appear in Exercises 31–36.

FIGURE 6.17

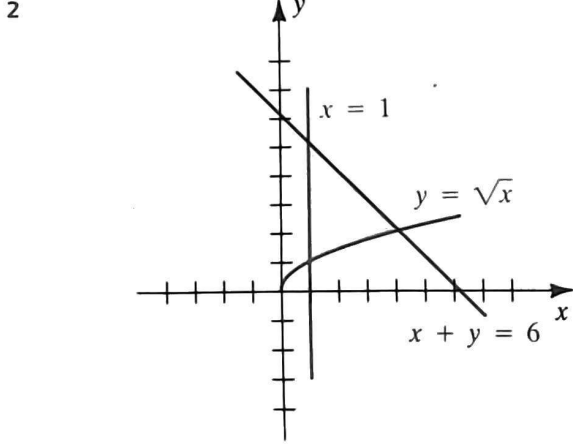
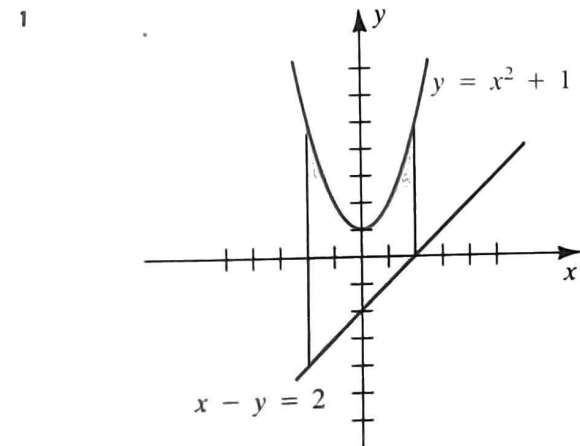
Stress-strain diagram for an elastic material



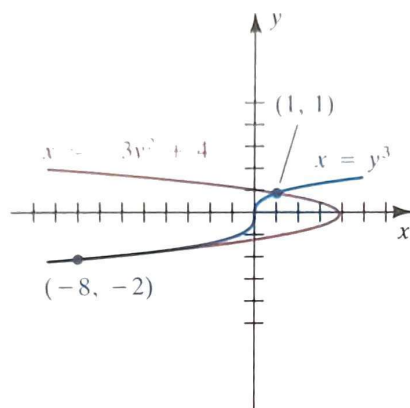
In scientific investigations, a physical quantity is often interpreted as an area. One illustration of this occurs in the *theory of elasticity*. To test the strength of a material, an investigator records values of strain that correspond to different loads (stresses). The sketch in Figure 6.17 is a typical stress-strain diagram for a sample of an elastic material, such as vulcanized rubber. (Note that stress values are assigned in the vertical direction.) Referring to the figure, we see that as the load applied to the material (the stress) increases, the strain (indicated by the arrows on the red graph) increases until the material is stretched to six times its original length. As the load decreases, the elastic material returns to its original length; however, the same graph is not retraced. Instead, the graph shown in blue is obtained. This phenomenon is called *elastic hysteresis*. (A similar occurrence takes place in the study of magnetic materials, where it is called *magnetic hysteresis*.) The two curves in the figure make up a *hysteresis loop* for the material. The area of the region enclosed by this loop is numerically equal to the energy dissipated within the elastic (or magnetic) material during the test. In the case of vulcanized rubber, the larger the area, the better the material is for absorbing vibrations.

EXERCISES 6.1

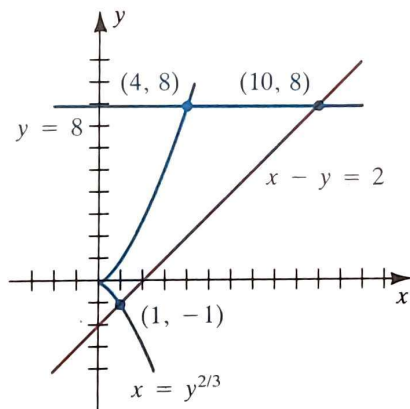
Exer. 1–4: Set up an integral that can be used to find the area of the shaded region.



3



4



Exer. 5–22: Sketch the region bounded by the graphs of the equations and find its area.

- | | |
|----------------------------|----------------------------------|
| 5 $y = x^2$; | $y = 4x$ |
| 6 $x + y = 3$; | $y + x^2 = 3$ |
| 7 $y = x^2 + 1$; | $y = 5$ |
| 8 $y = 4 - x^2$; | $y = -4$ |
| 9 $y = 1/x^2$; | $y = -x^2$; $x = 1$; $x = 2$ |
| 10 $y = x^3$; | $y = x^2$ |
| 11 $y^2 = -x$; | $x - y = 4$; $y = -1$; $y = 2$ |
| 12 $x = y^2$; | $y - x = 2$; $y = -2$; $y = 3$ |
| 13 $y^2 = 4 + x$; | $y^2 + x = 2$ |
| 14 $x = y^2$; | $x - y = 2$ |
| 15 $x = 4y - y^3$; | $x = 0$ |
| 16 $x = y^{2/3}$; | $x = y^2$ |
| 17 $y = x^3 - x$; | $y = 0$ |
| 18 $y = x^3 - x^2 - 6x$; | $y = 0$ |
| 19 $x = y^3 + 2y^2 - 3y$; | $x = 0$ |
| 20 $x = 9y - y^3$; | $x = 0$ |
| 21 $y = x\sqrt{4 - x^2}$; | $y = 0$ |
| 22 $y = x\sqrt{x^2 - 9}$; | $y = 0$; $x = 5$ |

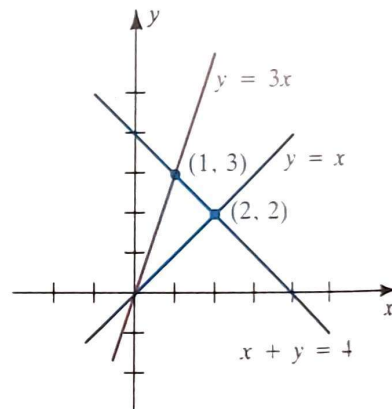
Exer. 23–24: Find the area of the region between the graphs of the two equations from $x = 0$ to $x = \pi$.

23 $y = \sin 4x$; $y = 1 + \cos \frac{1}{2}x$

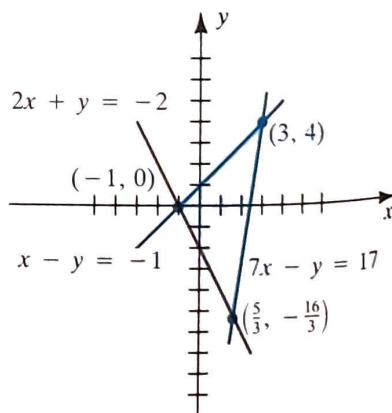
24 $y = 4 + \cos 2x$; $y = 3 \sin \frac{1}{2}x$

Exer. 25–26: Set up sums of integrals that can be used to find the area of the shaded region by integrating with respect to (a) x and (b) y .

25



26



Exer. 27–30: Set up sums of integrals that can be used to find the area of the region bounded by the graphs of the equations by integrating with respect to (a) x and (b) y .

27 $y = \sqrt{x}$; $y = -x$; $x = 1$; $x = 4$

28 $y = 1 - x^2$; $y = x - 1$

29 $y = x + 3$; $x = -y^2 + 3$

30 $x = y^2$; $x = 2y^2 - 4$

Exer. 31–36: Find the area of the region between the graphs of f and g if x is restricted to the given interval.

31 $f(x) = 6 - 3x^2$; $g(x) = 3x$; $[0, 2]$

32 $f(x) = x^2 - 4$; $g(x) = x + 2$; $[1, 4]$

33 $f(x) = x^3 - 4x + 2$; $g(x) = 2$; $[-1, 3]$

34 $f(x) = x^2$; $g(x) = x^3$; $[-1, 2]$

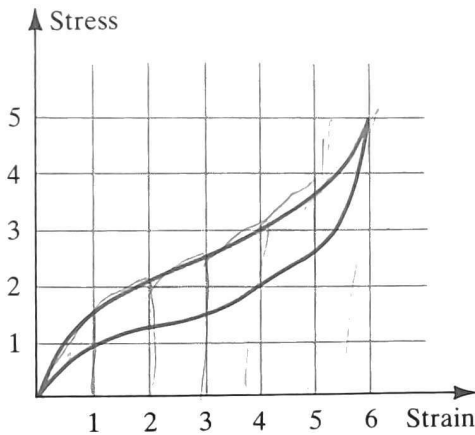
- 35 $f(x) = \sin x$; $g(x) = \cos x$; $[0, 2\pi]$
36 $f(x) = \sin x$; $g(x) = \frac{1}{2}$; $[0, \pi/2]$

Exer. 37–38: Let R be the region bounded by the graph of f and the x -axis, from $x = a$ to $x = b$. Set up a sum of integrals, not containing the absolute value symbol, that can be used to find the area of R .

- 37 $f(x) = |x^2 - 6x + 5|$; $a = 0$, $b = 7$
38 $f(x) = |-x^2 + 2x + 3|$; $a = -3$, $b = 4$

39 The shape of a particular stress-strain diagram is shown in the figure (see the last paragraph of this section).

EXERCISE 39



Estimate y -coordinates and approximate the area of the region enclosed by the hysteresis loop using, with $n = 6$,
(a) the trapezoidal rule
(b) Simpson's rule

- 40 Suppose the function values of f and g in the table below were obtained empirically. Assuming that f and g are continuous, approximate the area between their graphs from $x = 1$ to $x = 5$ using, with $n = 8$,
(a) the trapezoidal rule
(b) Simpson's rule

x	1	1.5	2	2.5	3	3.5	4	4.5	5
$f(x)$	3.5	2.5	3	4	3.5	2.5	2	2	3
$g(x)$	1.5	2	2	1.5	1	0.5	1	1.5	1

- [c] 41 Graph $f(x) = |x^3 - 0.7x^2 - 0.8x + 1.3|$ on $[-1.5, 1.5]$. Set up a sum of integrals not containing the absolute value symbol that can be used to approximate the area of the region bounded by the graph of f , the x -axis, and the lines $x = -1.5$ and $x = 1.5$.
[c] 42 Graph, on the same coordinate axes, $f(x) = \sin x$ and $g(x) = x^3 - x + 0.2$ for $-2 \leq x \leq 2$. Set up a sum of integrals that can be used to approximate the area of the region bounded by the graphs.

6.2 SOLIDS OF REVOLUTION

The volume of an object plays an important role in many problems in the physical sciences, such as finding centers of mass and moments of inertia. (These concepts will be discussed later in the text.) Since it is difficult to determine the volume of an irregularly shaped object, we shall begin with objects that have simple shapes. Included in this category are the solids of revolution discussed here and in the next section.

If a region in a plane is revolved about a line in the plane, the resulting solid is a **solid of revolution**, and we say that the solid is **generated** by the region. The line is an **axis of revolution**. In particular, if the R_x region shown in Figure 6.18(i), on the following page, is revolved about the x -axis, we obtain the solid illustrated in (ii) of the figure. As a special case, if f is a constant function, say $f(x) = k$, then the region is rectangular and the solid generated is a right circular cylinder. If the graph of f is a semi-circle with endpoints of a diameter at the points $(a, 0)$ and $(b, 0)$, then the solid of revolution is a sphere. If the region is a right triangle with base on the x -axis and two vertices at the points $(a, 0)$ and $(b, 0)$ with the right angle at one of these points, then a right circular cone is generated.

If a plane perpendicular to the x -axis intersects the solid shown in Figure 6.18(ii), a circular cross section is obtained. If, as indicated in the figure, the plane passes through the point on the axis with x -coordinate w , then the radius of the circle is $f(w)$, and hence its area is $\pi[f(w)]^2$. We shall arrive at a definition for the volume of such a solid of revolution by using Riemann sums.